

Protein Purification

Overview

To remove residual salts and concentrate the desired protein following large-scale production.

A. Materials

Item	Amount
Overnight liquid culture of His-tagged protein expression strain	As required (e.g., 50–1000 mL, induced)
Centrifuge (for cell harvest and supernatant clarification)	1
HisPur Ni-NTA Purification Kit	1 (choose spin column size)
10X PBS (200 mM sodium phosphate, 3M NaCl, pH 7.4)	Supplied in kit
2M Imidazole, pH 7.4	Supplied in kit
ddH ₂ O	As required
Centrifuge tubes (1.5, 15, or 50 mL)	Several
Benchtop centrifuge (for spin columns)	1

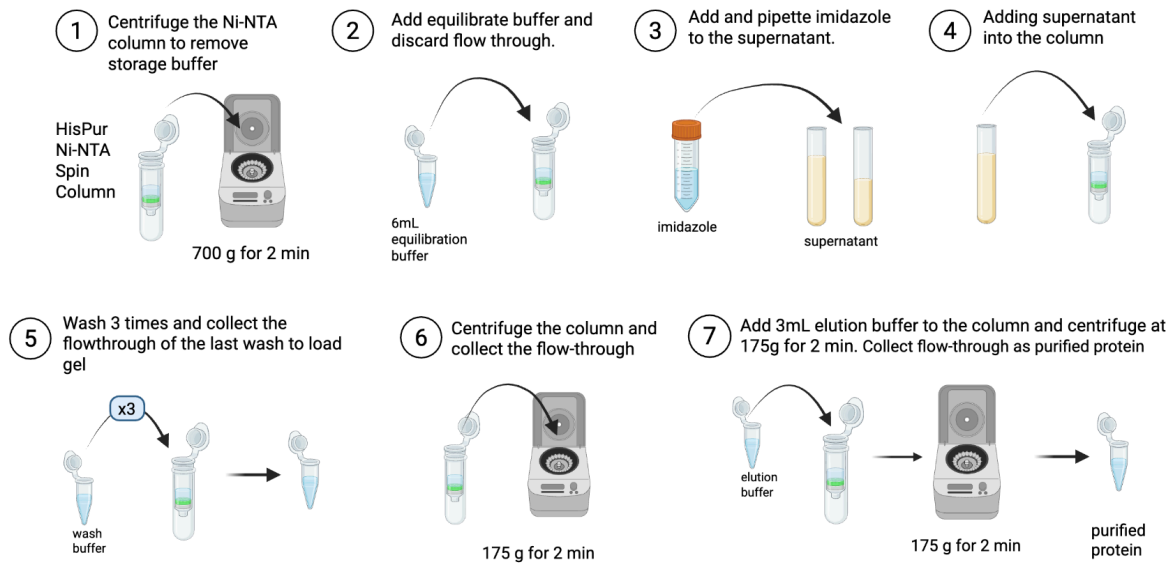
B. Procedure

1. Use the following buffer for the procedures:(formula for 50mL and buffers are prepared/stocked beforehand)

Buffer	Imidazole final concentration(m M)	10xPBS(mL)	2M imidazole(pM)	water(mL)
Equilibrate buffer	10	5	250	44.75
Wash buffer	25	5	625	44.375
Elute buffer	150	5	3750	41.25

2. Centrifuge the Ni-NTA column at 175g for 2 min to remove storage buffer
3. Add 6mL of equilibrate buffer
4. Before adding supernatant, add and pipette 237.5 μ L of imidazole to the tube of supernatant with 47.5mL volume and add 62.5 μ L to the tube of supernatant with 12.5mL volume
5. Adding all supernatant into the column by washing the walls(sides) of the column
6. Add 6mL of wash buffer and remove the flow-through. Repeat this step 3 times. *At the 3rd time, collect the flowthrough to load gel
7. Centrifuge at 175g for 2 min
8. Collect the flow-through from the last wash buffer step
9. Add 3mL of elute buffer
10. Centrifuge at 175g for 2 min
11. Collect flow-through as purified protein

Protein Purification



Protein Dialysis

A. Material

Item	Amount
Clarified protein extract	As prepared
4.8 L 1X PBS pH 8	1

Slide-A-Lyzer G3 Dialysis Cassettes, 10K MWCO, 3 mL (Thermo Fisher Scientific)	1
Ice bag	3-4
ddH ₂ O	As required
2L beaker	3
pH meter	1
Plastic wrap	1 box
Stir plate	1
Stir bar	1
Microtube(1.5mL)	several

B. Procedure

1. Dilute 10x PBS: Mix 160mL of 10xPBS and 1.6L of deionized water
2. Adjust pH to 8
3. Top up H₂O to 1.6L and wrap opening of beaker with plastic wrap
4. Prepare three beakers of 1.6L PBS
5. Open the dialysis cassette by gently twisting the cap counterclockwise
6. Add the protein lysate to the cassette, slowly withdrawing the pipette while dispensing. Don't overload the cassette
7. Remove the excess air in the cassette by simultaneously pressing the membrane gently on both sides using gloved thumb and forefinger and insert the cap
8. Insert cap and lock by gently twisting it clockwise until the arrows align and a firm stop is felt
9. Place dialysis cassette in 2 L beaker with 1.6 L 1X PBS and stir bar. Stir on ice for 2 hours
10. Replace 1X PBS and stir on ice for another 2 hours
11. Replace 1X PBS and stir on ice overnight
12. Transfer lysate containing recombinant protein from dialysis cassette to 1.5 mL microtubes
13. The recombinant protein is stored at -80°C

Protein Dialysis

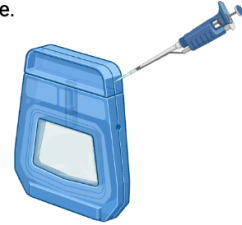
- ① Dilute 10X PBS by mixing 160mL of 10xPBS and 1.6L of deionized water



- ② Prepare three 1.6L 1X PBS



- ③ Add the sample to the cassette and remove the air inside.



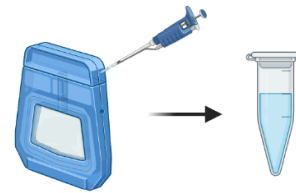
- ④ Place the cassette in the beaker with 1X PBS and a stir bar. Stir on ice for 2hrs.



- ⑤ Replace the 1X PBS and stir on ice for 2hrs then replace again overnight.



- ⑥ Transfer the recombinant protein from the cassette to the 1.5mL microtube.



Ni-NTA column Regeneration

A. Material

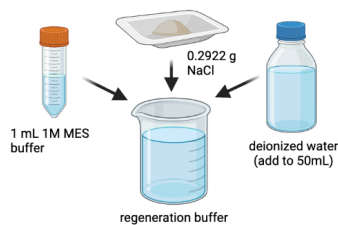
Item	Amount
1M MES buffer (pH 5.0) (Thermo Fisher Scientific)	1 mL
NaCl	0.2922 g
Deionized water	80mL
20% ethanol	15mL

B. Procedure

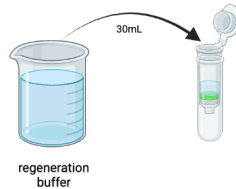
1. Prepare 50 mL regeneration buffer: Add 1 mL 1M MES buffer and 0.2922 g NaCl. Add deionized water to 50 mL.
2. Add 30mL of Ni-NTA regeneration buffer (10x of resin's volume) to the Ni-NTA column.
3. Discard the flow-through.
4. Add 30mL of deionized water (10x of resin's volume) to the Ni-NTA column to wash.
5. Discard the flow-through.
6. Add 15 mL of 20% ethanol and seal the Ni-NTA column.
7. Store the Ni-NTA column at 4 °C.

Ni-NTA Column Regeneration

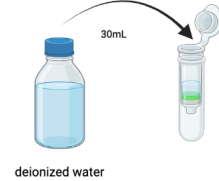
- ① Prepare 50mL regeneration buffer



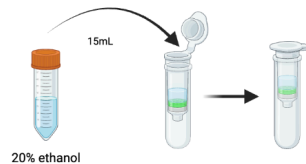
- ② Add 30mL of Ni-NTA regeneration buffer (10x of resin's volume) to the Ni-NTA column. Discard the flow-through.



- ③ Add 30mL of deionized water (10x of resin's volume) to the Ni-NTA column to wash. Discard the flow-through.



- ④ Add 15 mL of 20% ethanol and seal the Ni-NTA column.



- ⑤ Store the Ni-NTA column at 4°C

